

A model of technology diffusion in separate and unrelated dual markets

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Abstract

The main purpose of this paper is to propose a new model of technology diffusion for its implementation in separate and unrelated dual markets. Following a brief description of the classical models of diffusion in continuous markets and a study of dual market-based models, we propose a new model based on the conceptual principles of Moore (1991), which presuppose the existence of two independent adoption cycles in any process of technology diffusion. After a theoretical justification for the formulation of the model, its empirical validation is presented by comparing the predictive ability of the proposed model with that of the Bass mixed influence model (1969).

Keywords: Diffusion models, innovation, dual markets.

Introduction

The diffusion of an innovation is defined as the process by which it is transmitted over time among the members of a social system through certain communication channels (Rogers, 2003). While diffusion is affected by the type of innovation, communication channels (interpersonal and mass), social system and time, the notion refers essentially to a form of communication. Hence, diffusion is the spreading of messages related to new ideas embodied in innovations (products, processes and technologies) with the anticipation of a change in the behaviour of the receiver, expressed by the adoption or rejection of the innovation.

Since the publication of Bass' early works (1969), the literature on the dissemination of information, innovation or products has been extremely prolific. A considerable part of this literature reviews and validates the models proposed by different authors, while no less a significant part adds parameters to explain the basic models in the presence of market variables and economic variables (price, advertising, changes in GDP, etc.).

At the turn of the millennium, a line of research appeared in scientific marketing literature (Goldenberg, Libai and Muller, 2002; Golder and Tellis, 2004; Goldenberg, Libai, Muller and Peres, 2006) based on the idea that the diffusion of new products does not occur continuously in a single stage or market as assumed by most classical treatises on product life cycle. The studies on thirty product categories conducted by Golder and Tellis (2004) reveal regressions in the adoption curve in a significant number of cases. There is no explanation for these regressions if the adoption cycle is considered to affect a single market, nor do they fit in with most of the proposed models. Moore (1991) uses a conceptual approach, albeit without empirical evidence, to explain the existence of two separate markets in the cycle of technology adoption. The aim of this paper is to propose a mathematical model of innovation diffusion that incorporates the ideas of Moore regarding the two markets of the adoption cycle and allows empirical testing.

The paper is structured as follows: firstly, we present a review of the available literature on diffusion models in dual continuous markets. The new model is then developed from Bass' classical equation (1969), including descriptive parameters of the two markets and explaining the adoption of new products in more detail. Lastly, we conduct an empirical study comparing Bass' original model and the proposed model to determine the descriptive validity of the latter.

Diffusion models in continuous markets

Diffusion models have been developed to represent the form and speed of the diffusion of an innovation over time. In social sciences, diffusion models are typically based on mathematical formulas taken from physics and biology, specifically from models for the spread of epidemics (for example, see Bailey, 1957).

The pioneering models for innovation diffusion attempt to describe the changes in the behaviour of demand for a new product over time. Fourt and Woodlock (1960) explain it in terms of a coefficient of external influence, caused both by the individual's intrinsic tendency to adopt and by the influence of external factors such as mass media. (Ruiz and Mas, 2001). In this early model, the diffusion rate depends exclusively on the number of potential adopters $[M - N(t)]^1$, with the number of previous adopters of the innovation having no effect on the adoption process. The potential adopters are influenced exclusively by external sources (advertising, product information, etc.).

The mathematical model of external influence is:

$$\frac{dN(t)}{dt} = a[M - N(t)] \quad (1)$$

In another pioneering model, Mansfield (1961) proposes a factor of internal influence or imitation due to personal contact with previous adopters. This early internal influence model considers that there is no intrinsic tendency to adopt in individuals and that external sources of information are not significant in the diffusion process. Unlike the previous model, it determines that communication between the members of the social system is the only factor that influences the diffusion process. Hence, the rate of diffusion is proportional both to the number of potential adopters $[M - N(t)]$ and the number of previous adopters $N(t)$.

The mathematical model of internal influence is:

$$\frac{dN(t)}{dt} = bN(t)[M - N(t)] \quad (2)$$

¹

M is the potential market of adopters.

$N(t)$ represents the cumulative number of individuals who have adopted the technology up to t .

a is the coefficient of external influence.

b is the coefficient of internal influence.

Both elements are considered simultaneously in the Bass model (1969) of mixed influence, which, as the name suggests, considers that the adoption process is generated by the influence of both internal and external variables. Hence, the mathematical formula for this model is the sum of the previous two.

$$\frac{dN(t)}{dt} = (a + bN(t))[M - N(t)] \quad (3)$$

Since these models were proposed, diverse variables have been incorporated into the formulation of the equations, especially those relating to marketing decisions, such as price or advertising. Table 1 contains a non-exhaustive list of diffusion models that incorporate marketing variables.

Author	Marketing Variables	Equation	Parameters
Bass (1969)	N/A	$\frac{dF(t)}{dt} = p + (q - p)F(t) - qF(t)^2$	p = Coefficient of innovation q = Coefficient of imitation
Robinson and Lakhani (1975)	Price	$\frac{dF(t)}{dt} = [p + (q - p)F(t) - qF(t)^2] \cdot e^{-k \text{Pr}(t)}$	$\text{Pr}(t)$ = Price function k = Coefficient p = Coefficient of innovation q = Coefficient of imitation
Bass (1980)	Price	$\frac{dF(t)}{dt} = [p + (q - p)F(t) - qF(t)^2] \cdot c \cdot [\text{Pr}(t)]^2$	$\text{Pr}(t)$ = Price q = Price elasticity c = Cost
Kalish (1985)	Price and advertising	$\frac{dI}{dt} = [1 - I] \cdot \left[f(A(t)) + bI + b' \frac{Y(t)}{m} \right]$ $\frac{dY}{dt} = \left[g \left(\frac{\text{Pr}(t)}{uY(t)/m} \right) \cdot I - Y(t) \right] \cdot k$	$Y(t)$ = Cumulative sales to time t $\text{Pr}(t)$ = Price function at time t m = Market potential $A(t)$ = Advertising function at time t I = Level of information g and u are b, b', k = Parameters
Horsky and Simon (1983)	Advertising	$S(t) = [\alpha + \beta \ln(A(t)) + qY(t-1)][m - Y(t)]$	$A(t)$ = Advertising function at time t β = Advertising effectiveness $Y(t)$ = Cumulative sales at time t $S(t)$ = Sales at time t m = Market potential
Kamakura and Balasubramanian (1988)	Price	$S(t) = [p + qY(t)] \text{Pr}(t)^{\beta_1} [\Theta M(t) \text{Pr}(t)^{\beta_2} - Y(t)]$	$Y(t)$ = Cumulative sales at time t $S(t)$ = Sales at time t $\text{Pr}(t)$ = Price function at time t $M(t)$ = Population function at time t Θ = Ultimate penetration level β_1 = Impact of price on adoption speed β_2 = Impact of price on market potential p = Coefficient of innovation q = Coefficient of imitation
Horsky (1990)	Price	$S(t) = \left[\frac{\theta M(t)}{1 + \exp \left\{ -\frac{K + w(t) - k \text{Pr}(t)}{\delta(t)} \right\}} - Y(t) \right] [p + qY(t)]$	$w(t)$ = Average wage rate of the population $\text{Pr}(t)$ = Average price of the product δ = Dispersion of both distributions K = Time-saving attribute of the new product k = Utility-enhancing attribute of the new product $Y(t)$ = Cumulative sales up to time t $S(t)$ = Sales at time t $M(t)$ = Population at time t d = Fraction of potential buyers in the population p = Coefficient of innovation q = Coefficient of imitation
Jain and Rao (1990)	Price	$S(t) = [m \text{Pr}(t)^{-\eta} - Y(t-1)] \frac{F(t) - F(t-1)}{1 - F(t-1)}$ $S(t) = [m - Y(t-1)] \text{Pr}(t)^{-\eta} \frac{F(t) - F(t-1)}{1 - F(t-1)}$	$Y(t)$ = Cumulative sales up to time t $S(t)$ = Sales at time t $\text{Pr}(t)$ = Price function at time t q = Price elasticity

Bass, Jain and Krishnan (1994)	Price and advertising	$\frac{dF(t)}{dt} = [p + (q - p)mF(t) - qm(F(t))^2]x(t)$ where $x(t) = 1 + \beta_1 \frac{Pr'(t)}{Pr(t)} + \beta_2 \frac{A'(t)}{A(t)}$	m = Market potential $Pr(t)$ = Price function at time t $Pr'(t)$ = Ratio of price change at time t $A(t)$ = Advertising function at time t $A'(t)$ = Ratio of advertising change at time t β_1 = Coefficient of price effectiveness β_2 = Coefficient of advertising effectiveness p = Coefficient of innovation q = Coefficient of imitation
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Table 1. Diffusion models in continuous markets with marketing variables. (Adapted from Radas, 2005)

Diffusion models in dual markets

In the previous cases, the diffusion models are based on the premise of the existence of a single continuous market that increases over time. This assumption remained unchanged for nearly three decades while the vast majority of researchers on the subject were developing models to explain the influence of marketing variables on adoption cycles, without questioning the premise of a single market.

Only a fraction of the literature up to the early 1990s proposes new models identifying two population subgroups based on propensity to adopt new products or technologies, each of which is modelled using different equations. Along these lines, Tanny and Derkzo (1988) propose dividing the population into innovators and imitators (Figure 1). Their model indicates that innovators purchase the product exclusively in response to external sources (advertising, product communication or any source of information that does not come from another user), while imitators adopt in response both to external sources and the effect of interaction with previous adopters.

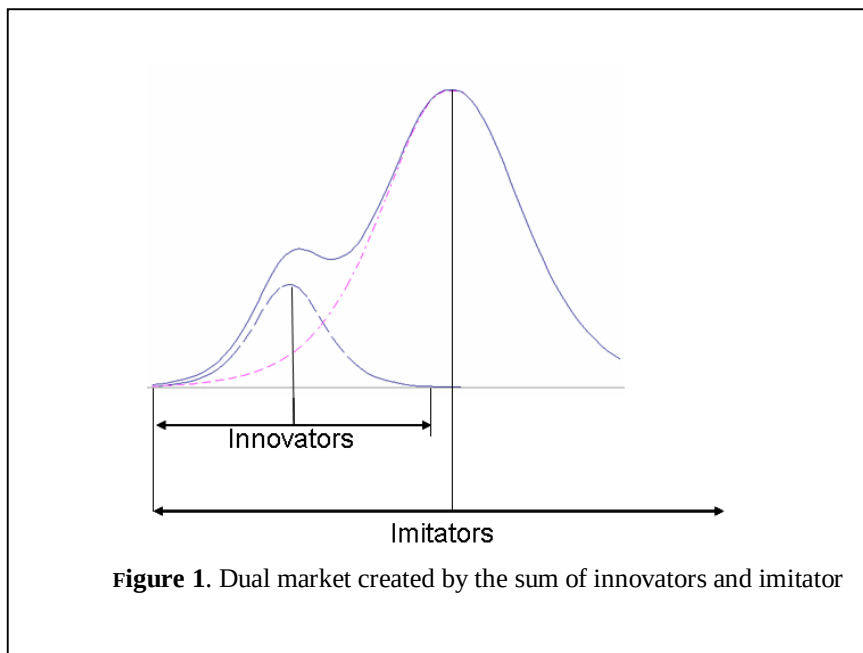


Figure 1. Dual market created by the sum of innovators and imitator

In mathematical terms, the model is described as follows. The equation for the innovators market is:

$$\dot{x}_1 = p_1 (n_1 - x_1) \quad (4)$$

while for the imitators market it is:

$$\dot{x}_2 = (p_2 + q_2 (x_1 + x_2)/n) (n_2 - x_2) \quad (5)$$

In the equation governing the first market (4), there is only one coefficient of external influence (p_1), which is equivalent to the Bass equation if the coefficient of external influence (q) is equal to 0. In the equation modelling the second market (5), the term $q_2 (x_1 + x_2)/n$ ($n_2 - x_2$) is the contribution from x_2 to the rate of change due to the fraction $(x_1 + x_2)/n$ of the population that has already adopted at time t , while the term $p_2 (n_2 - x_2)$ models the variation of x_2 due to the innovators in the second market. In this first dual market model the empirical results were not satisfactory. Convergence was not achieved in any of the eleven product categories analysed (Tanny and Derkzo, 1988).

In a later model, Steffens and Murthy (1992) divided the two market segments according to the product attributes defined by Nelson (1970) and Darby and Karni (1973). These authors divide products into "search goods" and "experience goods". Search products have intrinsic and extrinsic elements that can be objectively evaluated based on the information provided by the vendor. With experience goods, however, the information provided about them is insufficient to decide the purchase. Consumers can only evaluate the suitability of the goods by personally examining, testing or consuming them (Rodríguez, 2008).

This duality led Steffens and Murthy to propose two types of population based on their propensity to buy search or experience goods, calling the first group innovators and the second, imitators. Innovators are willing to adopt based on the search attributes of the product, which are available through advertising sources, manufacturer's specifications or direct information at the point of sale. Imitators delay their purchase until they are informed of the experience attributes of the product. This information is assumed to come exclusively from a previous adopter, since the latter is the only person who can convey experience with the product.

Following this logic, the equation governing the first market contains coefficients of internal and external influence:

$$\frac{dN_1}{dt} = (K_1 - N_1(t))(\alpha + \beta N_1(t)) \quad (6)$$

while the second only contains the coefficient of internal influence:

$$\frac{dN_2}{dt} = (K_2 - N_2(t))\gamma(N_1(t) + N_2(t)) \quad (7)$$

The influence of the first market on the second is modelled with the term:

$$\gamma(N_1(t) + N_2(t))$$

The idea of dual markets was not taken up again until Goldenberg, Libai and Muller (2002) found that between a third and a half of the sales of innovative products in the consumer electronics industry followed the same pattern. Their study showed that after an initial period of increased sales, there was a clear recession that remained for some time until sales increased again, this time above the initial peak. They call this new pattern "saddle" in clear reference to the shape of the sales curve (Figure 2).

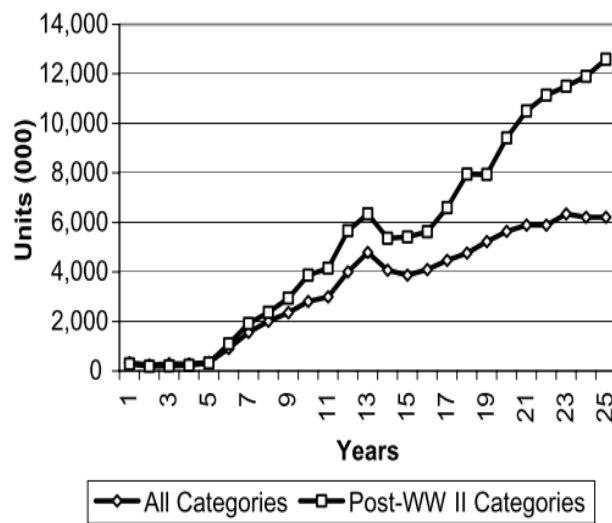
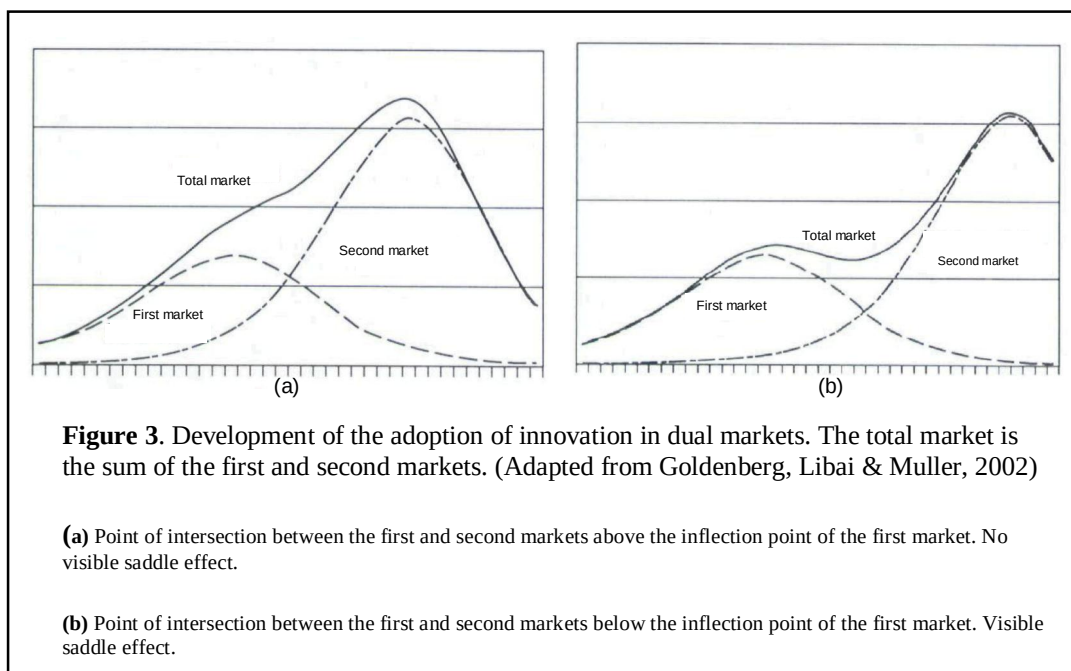


Figure 2. Saddle effect on new consumer durable.
Golder & Tellis (2004)

The authors conclude that the saddle effect can only be explained by the existence of two markets that have a time gap between them and adopt the product at different speeds. If there is a pronounced time difference between the two markets, the saddle effect will be observed. If we draw different adoption curves for each market, we notice that, as the curve of the second market is shifted along the time axis, the sum of the two (the total market) takes on different forms (Figure 3). If the curve of the second market intersects the curve of the first one below its inflection point, the saddle effect will be visible for the total market; otherwise, it will grow continuously and the saddle effect will not be significant.



The next contribution to diffusion based on dual markets was the model proposed by Muller and Yogev (2006). This model, unlike the previous ones, no longer considers two separate markets of innovators and imitators. Following the postulates of Moore (1991), the authors consider that both the first and the second markets contain innovators and imitators, so both markets should contain coefficients of internal and external influence. Their model considers that the two markets communicate with one another, i.e. some of the adopters from the second market are the result of the influence of the members of the first market.

According to this hypothesis, the expression of the first market in the model of Muller and Yogev (2006) is the equation of Bass (3), where p_i is the coefficient of external influence, q_i is the coefficient of internal influence and N_i is the market potential of the first market.

$I(t)$ is the number of adopters in the period t :

$$\frac{dI(t)}{dt} = (p_i + q_i \frac{I(t)}{N_i})(N_i - I(t)) \quad (8)$$

For the second market, the coefficient of internal influence is split in two: q_m and q_{im} . The first regulates the effect of imitation in the first market while the second models the communication between the first and second markets.

The coefficient N_m is the market potential of the second market while $M(t)$ represents the number of adopters in the period t .

$$\frac{dM(t)}{dt} = (p_m + q_m \frac{M(t)}{N_i + N_m} + q_{im} \frac{I(t)}{N_i + N_m})(N_m - M(t)) \quad (9)$$

The sum of these two equations describes the adoption cycle, showing the dual market effect.

In the experimental validation carried out by Muller and Yogev (2006) using the above equations, several problems arose that are described by the authors. Firstly, estimating a model with seven parameters causes instability and convergence problems. The estimation is performed by assigning $p_m = 0$ and setting a value of N_m obtained or estimated from external sources. Thus, we can reduce the number of parameters to five and use methods of non-linear regression to obtain the coefficients. To obtain final data, however, a priori estimates of market potential are always required, and this is precisely one of the variables that the model has to provide. As a result, the predictive power of this model will always be limited to the existence of data allowing us to estimate a priori the values of the first potential market.

A final contribution to the study of dual markets (Van den Bulte and Joshi, 2007) presents two markets defined by the following probability functions:

$$f_1 = (p_1 + q_1 F_1(t)) (1 - F_1(t)) \quad (10)$$

$$f_2 = (q_2 [w F_1(t) + (1-w) F_2(t)]) (1 - F_2(t)) \quad (11)$$

The authors call the first market influencers and the second, imitators.

To summarise the model, it proposes two types of markets, the first of which contains coefficients of internal and external influence, while the second only contains coefficients of internal influence, i.e. in the second market, the authors

only consider the possibility that the product is purchased by imitation. Nonetheless, this process of imitation is influenced by the number of adopters in the first market. Hence, the term $F_1(t)$ appears in both expressions, showing that the number of adopters in the second equation depends directly on the number of adopters in the first market. The coefficient w indicates the relative importance of the influencers on the behaviour of the imitators.

Table 2 summarises the dual market models based on their use of coefficients of internal or external influence in each of the two markets.

Author	Model	Coefficients of influence
Tanny and Derkzo (1988)	$x_1 \dot{\theta} = p_1 (n_1 - x_1)$ $x_2 \dot{\theta} = (p_2 + q_2 (x_1 + x_2)/n) (n_2 - x_2)$	External Internal and external
Steffens and Murthy (1992)	$\frac{dN_1}{dt} = (K_1 - N_1(t))(\alpha + \beta N_1(t))$ $\frac{dN_2}{dt} = (K_2 - N_2(t)) \gamma (N_1(t) + N_2(t))$	Internal and external Internal
Muller and Yogev (2006)	$\frac{dI(t)}{dt} = (p_i + q_i \frac{I(t)}{N_i})(N_i - I(t))$ $\frac{dM(t)}{dt} = (p_m + q_m \frac{M(t)}{N_i + N_m} + q_{im} \frac{I(t)}{N_i + N_m})(N_m - M(t))$	Internal and external Internal and external
Van den Bulte and Joshi (2007)	$f_1 = (p_1 + q_1 F_1(t)) (1 - F_1(t))$ $f_2 = (q_2 [w F_1(t) + (1-w) F_2(t)]) (1 - F_2(t))$	Internal and external Internal

Table 2. Dual-market diffusion models

Conceptual framework of dual markets

Golder and Tellis (2004) offer an explanation for the existence of two markets based on the theory of informational cascades (Bikchandani, Hirshleifer and Welch, 1992). Cascades occur when many consumers base their decision on the behaviour of the few, instead of basing it on individual assessments of the usefulness of the alternatives. A handful of consumers choose to buy a new product on its own merits and then a group of consumers buys the product in imitation of the earlier group, causing an increase in sales. This sales growth acts as a catalyst and encourages new consumers to buy the new products in what could be termed a

"positive cascade". This causes sales to rise above what would be expected if they were solely due to private individual assessments of the usefulness of the product by consumers. The cascades formed in this type of process are fragile. Slight market turbulence can cause a downturn in sales, triggering a negative cascade. Such behaviour may be the reason for the drop in sales of a new product after its initial phase and could explain the rebound in sales after the turbulence.

Another explanation for the phenomenon, also offered by Golder and Tellis (2004), suggests that the recession is caused by consumption power criteria. Thus, a decline in national income or a macroeconomic contraction may trigger a decline in propensity to consume, which would result in a drop in sales of non-essential new products (Deleersnyder et al., 2004).

Moore (1991) proposed a conceptual analysis of the dual-market phenomenon long before it was empirically tested by Goldenberg, Libai and Muller (2002). His model describes how, during the early phase of the adoption cycle, decisions to purchase the product are dominated by tech-savvy enthusiasts and visionaries² representing 15% of the total potential market (Rogers, 2003). Their decisions to adopt are based exclusively on technological values and product specifications. In contrast, for the next group of consumers in the adoption cycle – the most important one³ since it constitutes 78% of the potential market – adoption decisions are not based on the formal or functional characteristics of the product, since the prevailing factors in the purchase decision are the existence of standards, support infrastructure, competition and the reliability offered by the manufacturers. Lastly, the members of the last group of the adoption cycle⁴ will purchase products reactively and only when they reach maturity, when prices are low enough and they can consider the product or technology a commodity.

The motivations behind adoption among members of the first and second markets are radically different: while the interest of the members of the first market lies with the characteristics of the product (speed, usability, functionality), the members of the second market are influenced by the characteristics of the market (installed base, third parties, standards, technical support, warranties). In short, the rate of adoption is the result of feedback between the two groups. The members of the second market will not buy the product until there is a sufficient critical mass of adopters who, logically, can only be members of the first market.

Nonetheless, critical mass is not the only decisive factor in the adoption of the product. Standards, competition and brand also play a role. Thus, although the critical mass has to be created by the members of the first market, the industry must often drive the transition between the two markets by providing the necessary elements to motivate individuals in the second market to adopt according to their adoption criteria.

² Innovators (2.5%) and early adopters (13.5%) in Rogers' categorisation.

³ Early majority (34%) and late majority (34%) in Rogers' categorisation.

⁴ Laggards (16%) in Rogers' categorisation.

The lack of a critical mass or the lack of resources in the industry to lay the foundations for a market would explain the failure of a large group of new products to reach the second market. Similarly, the lack of time synchronisation between the two markets would account for the "saddle" effect in the adoption curve.

In Moore's model, the channels of communication between the two markets are vertical, i.e. communication and influence only takes place between the members of each market. The markets are also unrelated: although horizontal communication is established between the members of the first and second markets, it has no influence on the adoption cycle because the motivations to purchase the product in the two markets are different.

Description of the proposed model

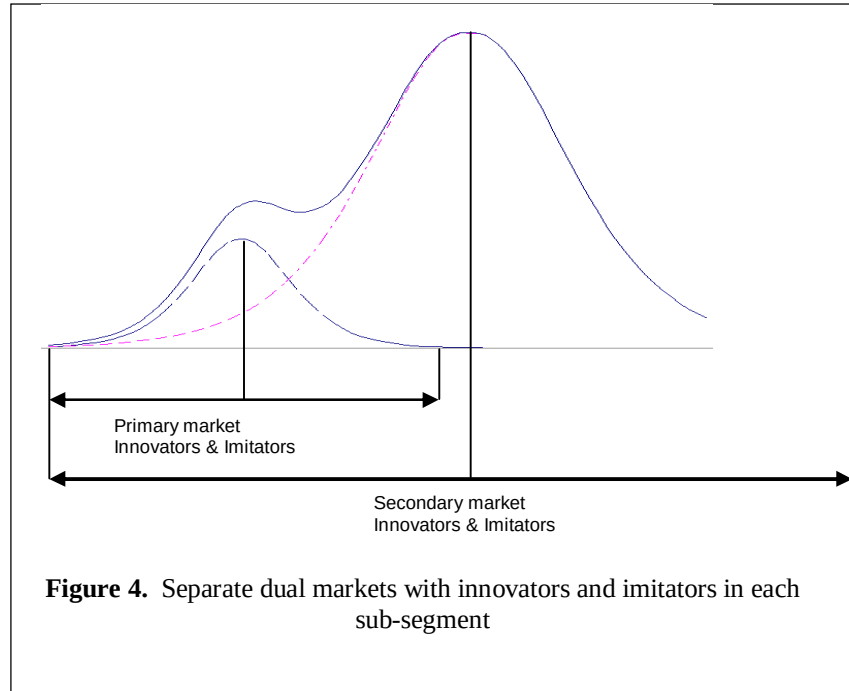
The dual market models presented in the previous sections have one criterion in common: the intercommunication and influence between the first and second markets. For example, in the proposals of Tanny and Derkzo (1988), Steffens and Murthy (1992) and Van den Bulte (2007), communication between the two markets is absolutely necessary because their models are based on the innovators/imitators dichotomy and the diffusion process must go through the first stage in order for the second to develop. In other words, if there is zero communication between the two markets, the diffusion process cannot continue and the adoption cycle is limited to the first market. In all of the above cases, if the first market were null, the second would not exist. It is assumed that, although the two markets are different, there has to be some sort of interaction between them to explain the influence of the adopters in the first group on the second.

In line with the conceptual framework of Moore (1991), the communication between the two markets in the proposed model is not relevant. Furthermore, although the first market is needed in the adoption cycle in order to develop the second, for practical purposes, we can consider the two markets as being virtually independent with their own coefficients of internal and external influence.

The proposed model makes the following assumptions:

1. Existence of a dual market: the first market will consist of innovators and early adopters while the second will contain all other segments of the adoption cycle described by Rogers (1985).
2. Each market includes innovative and imitators, the sum of which is the total market. There are independent coefficients of internal and external influence in each case (Figure 4).

3. Although there is a flow of information between the first and second markets, this is not relevant to the decision to adopt technologies made by the members of each market.
4. The flow of information is relevant only if it is *intra market*. This is critical because all studies of dual markets consider part of the second market to be constructed through the influence of the first. In this paper, we intend to treat the study of dual markets as two virtually independent and hence confined markets.
5. It is considered that the reasons for adopting the product are different in the two markets.



To model each of the separate markets, we have opted for the mixed influence model described above, whose equation is:

$$\frac{dN(t)}{dt} = (a + bN(t))[M - N(t)]$$

The choice of this model a priori is justified by its simple formulation, its well-known mathematical properties and its exact analytical solution. Hence, the adjustment of the function to the data is relatively straightforward. Moreover, in earlier works by several different authors, the mixed diffusion model developed by Bass has attained a considerable fit.

Since we can consider the diffusion process to be the sum of two separate markets, the expression for the first market based on (3) is:

$$f_1(t) = \frac{dF_1(t)}{dt} = (p_1 + q_1 F_1(t))(M_1 - F_1(t)) \quad (12)$$

whose analytical solution is:

$$F_1(t) = \frac{1 - e^{-(p_1 + q_1)t}}{1 + \left(\frac{q_1}{p_1}\right)e^{-(p_1 + q_1)t}} \quad (13)$$

Assuming $F_1(t) = N_1(t)/M_1$ where $F_1(t)$ is the fraction of potential adopters who adopt the product by time t , the cumulated number of adopters will be:

$$N_1(t) = M_1 F_1(t) = M_1 \frac{1 - e^{-(p_1 + q_1)t}}{1 + \left(\frac{q_1}{p_1}\right)e^{-(p_1 + q_1)t}} \quad (14)$$

For the second market, it is:

$$f_2(t) = \frac{dF_2(t)}{dt} = (p_2 + q_2 F_2(t))(M_2 - F_2(t)) \quad (15)$$

whose analytical solution for cumulated adopters is:

$$N_2(t) = M_2 \frac{1 - e^{-(p_2 + q_2)t}}{1 + \left(\frac{q_2}{p_2}\right)e^{-(p_2 + q_2)t}} \quad (16)$$

The expression for the total market is the sum of the two separate markets:

$$\begin{aligned} x(t) &= f_1(t) + f_2(t) = \frac{dF_1(t)}{dt} + \frac{dF_2(t)}{dt} = \\ &= (p_1 + q_1 N_1(t))(M_1 - N_1(t)) + (p_2 + q_2 N_2(t))(M_2 - N_2(t)) \end{aligned} \quad (17)$$

The solution to the above differential equation for cumulated number of adopters is:

$$N(t) = N_1(t) + N_2(t) = M_1 \frac{1 - e^{-(p_1 + q_1)t}}{1 + \left(\frac{q_1}{p_1}\right)e^{-(p_1 + q_1)t}} + M_2 \frac{1 - e^{-(p_2 + q_2)t}}{1 + \left(\frac{q_2}{p_2}\right)e^{-(p_2 + q_2)t}} \quad (18)$$

Where p_1 and q_1 are the coefficients of external and internal influence, respectively, on the first market, whose potential is M_1 , while p_2 and q_2 express the coefficients of external and internal influence for the second market, whose potential is M_2 .

Thus, this six-parameter model considers the two markets separately, in accordance with the conceptual work developed by Moore (1991) and in contrast to the models discussed earlier that propose the interaction of the first market with the second. The model we have described allows us to obtain the coefficients directly using non-linear regression programs.

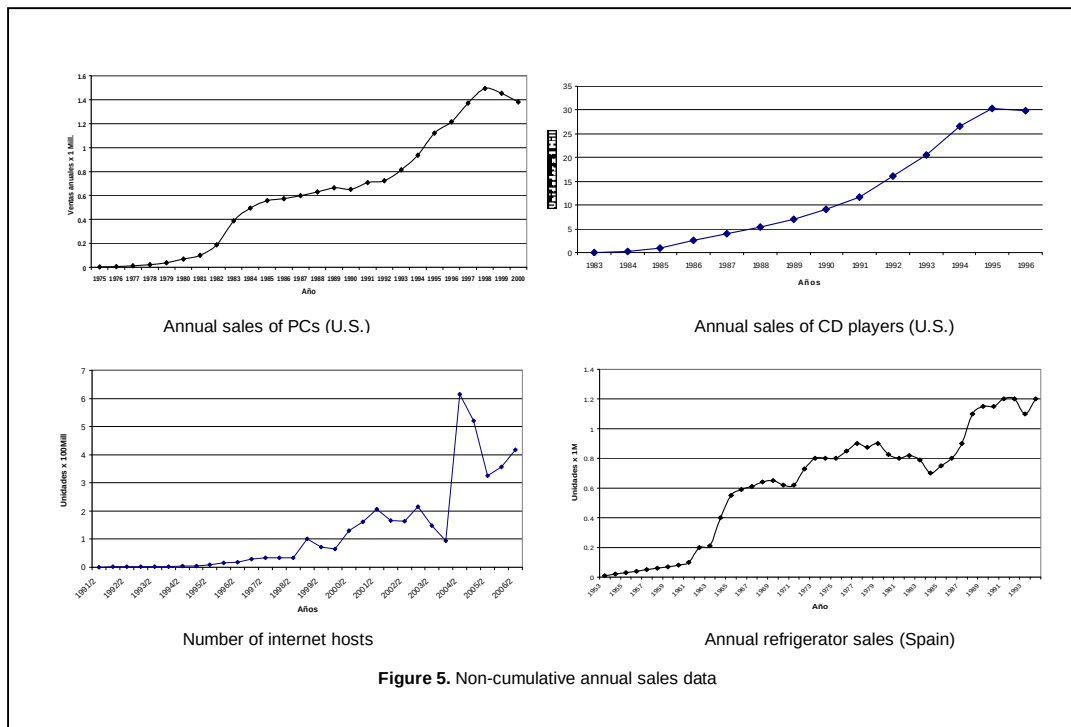
Validation of the model

Having described the model, the next step is its validation using empirical data. We will do this using temporal data on the sales of various consumer electronics goods. The data sets to be analysed reveal different "saddle" shapes in the total number of adopters, indicating the possible existence of a dual market in their adoption cycle. The next step in the validation of the model will be to obtain the coefficients that describe the adoption curves, both for the model described and for the Bass model. To do this, we will use the non-linear regression method described below. Lastly, the results obtained with the Bass model and the proposed model will be compared to determine their ability to describe the diffusion process.

Market data

The temporal data relate to four different cases: non-aggregate annual sales of computers in the U.S., those of CD players in the U.S., those of refrigerators in Spain and the emergence of Internet hosts (Figure 5). In the first two cases, the data source is <http://www.bassbasement.org>, while the data on refrigerators in Spain is taken from the study by Tellis, Stremersch and Yin (2003) and the data on Internet hosts is obtained from the work of López, Orero and Arroyo (2007).

Since a non-linear regression method is used to estimate the coefficients, the temporal data includes at least twelve periods.⁵ Hence, given that the average value of the first adoption peak is 13 years, we have opted for an annual time scale that allows us to show the different areas described above on a graph.



Estimation of coefficients

Bass and most authors of early studies on diffusion use an OLS (Ordinary Least Squares) estimation to calculate the values of the coefficients of innovation, imitation and market (p , q , M), which gives reasonable results with few data samples. To improve the estimation of the parameters, Schmittlein and Mahajan (1982) introduced the MLE (Maximum Likelihood Estimation) method and

⁵ Below this value, it is considered that other methods, such as MLE, can deliver better results. Srinivasan and Mason (1986).

Srinivasan and Mason (1986) subsequently developed the non-linear regression estimation model. (NLLS). In contrast to OLS, these two techniques have the advantage of not introducing bias to the time intervals and allow us to obtain standard errors of the parameters p , q and M . In their comparison of the two models, Putsis and Srinivasan (2000) conclude that there is evidence of the superiority of non-linear regression method compared to MLE.

The non-linear estimation method is also the main method used by researchers, especially since it is possible to use mathematical programs that automatically obtain the coefficients using an iterative method. As pointed out by Sánchez González (1999), most non-linear models obtain parameters as an optimisation of some objective function, usually the residual sum of squares. Hence, we calculate the values of the parameters that minimise the sum of squares of the residuals, i.e. the vertical distance between the forecasted and actual values.

In this paper, we have chosen to use the NREG program,⁶ which uses the above principle to calculate the coefficients. Its C-like programming language allows for quick and easy entry of the functions and parameters to be estimated.

The program will be used to obtain the coefficients for both the Bass model and the modified model.

```

Title: "Modified Bass estimation coefficients";
Variable t; // Time
Variable num; // Number of new adopters
Parameter p,q,r,s;
Parameter M,N;
Function num =(M*(1-(exp(-t*(p+q)))) / (1+((q/p)*exp(-t*(p+q)))))+(N*(1-(EXP(-t*(s+r)))) / (1+((s/r)*EXP(-t*(s+r)))));
plot;
Data;
1 0.00916
2 0.034021
3 0.068183
4 0.12309
5 0.20694
6 0.35089
7 0.52831
8 0.75571
9 1.1033
10 1.6009
11 2.4134
12 3.3786
13 4.4043
14 5.5312
15 6.9209
16 8.6047
17 10.948
18 12.837
19 14.077

```

Table 3. Program for calculating coefficients in the NREG editor

⁶ NLREG is a statistical analysis program that performs linear and non-linear, surface and curve-fitting regression analysis. The program determines the values of the parameters for an equation, which means that the equation provides a better fit for a set of data values and handles linear, polynomial, exponential, logistic, periodic and non-linear functions in general. NLREG can handle virtually any function whose form can be expressed algebraically.

The values obtained by calculating the diffusion coefficients using the proposed model and the Bass model are shown in Table 4.

	PCs (U.S.)	CD players (U.S.)	Internet hosts	Refrigerators (ES)
Proposed model				
p1	0.000767	0.0039	3.1064E-10	0.0001368
q1	0.545	0.8634	0.7385	0.17203
p2	0.000254	0.0006	0.0002	0.0018357
q2	0.29	0.5178	0.3494	0.23244
M1	3.7	1.5914	24.8090	27.58368
M2	20.017	23.6117	17.3653	12.373
Bass				
p	0.001975	0.002	0.000226	0.002665
q	0.1798	0.394	0.210	0.11476
M	29.247	31.414	84.69	33.86

Table 4. Coefficients of the proposed and Bass diffusion models

Results

In the first comparative analysis (Table 5) of the two models taking into account the adjusted R^2 value, the proposed model gives better results than the Bass model in all cases.

	PC's (USA)	CD Players (USA)	Host Internet	Refrigerators(ESP)
Proposed Model				
R2	99.99%	100.00%	99.82%	99.96%
Bass				
R2	99.66%	99.97%	99.63%	99.52%

Table 5. Comparison of R^2 . Proposed model compared to Bass model

If the two models had the same number of coefficients, this value alone would justify the goodness and superiority of the proposed model. However, as pointed

out by Motulsky and Christopoulos (2003), in non-linear regression, non-linear models should not be compared using the R^2 coefficient. As they are more flexible, models with more parameters will provide a better fit than those with fewer parameters, so this effect should be taken into account when comparing models. Given that our models are nested, an initial comparison of the two will be made on mean absolute percentage error (MAPE⁷). The MAPE data obtained in the proposed model are superior to those obtained in the classical Bass model.

	PC's (U SA)	CD Players (U SA)	Host Internet	Refrigerators (ESP)
Proposed model				
MAPE	0.23	0.7380	0.22	0.22
Bass				
MAPE	2.04	1.834	0.59	0.94

Table 6. MAPE comparison. Proposed model compared to Bass model

To compare MAPE results, the comparison of the results is complemented by the Akaike information criterion (AIC) (Akaike, 1974). This is a comparison model based on information theory for comparing both related and unrelated models.

The Akaike model is defined by the equation (19), where N is the number of data in the series, K is the number of parameters plus one and SSreg is the sum of squares of the vertical distances of the forecasted points in comparison to the data curve.

$$AIC = N \cdot \ln(SSreg / N) + 2 \cdot K \quad (19)$$

For models where the number of data N is small in comparison to the number of parameters K, a corrected version of the above formula is used:

$$AIC = N \cdot \ln(SSreg / N) + 2 \cdot K(K+1) / (N-K-1) \quad (20)$$

⁷ Mean Absolute Percentage Error. Statistic used to measure the quality of fit in the sample. A_t is the actual value, F_t is the forecast value and n is the number of samples.

$$MAPE = \frac{1}{n} \sum_{t=1}^n \left| \frac{A_t - F_t}{A_t} \right|$$

	PC's (U SA)	CD Players (U SA)	Host Internet	Refrigerator (ESP)
Proposed Model				
AIC	-137.53	-67.74	-28.77	-137.22
Bass				
AIC	-54.37	-66.87	-23.71	-36.11

Table 7. AIC comparison. Proposed model compared to Bass model

The AIC values obtained are dimensionless and can have both positive and negative signs. The AIC result has no significant value in itself; rather, the comparison of the AIC values of the models under study determines which of the models reveals a better fit in relation to the original values (Table 7).

$$\Delta AIC = AIC_B - AIC_A \quad (21)$$

Intuitively, if AICA is the model with fewer parameters, it will have a worse fit than the model with more parameters. Hence, the values of SSA will be greater than those of SSB. Since the logarithm of a fraction is always negative, the first term of the equation will be negative. The second term of the equation relates the coefficients. Since KA is greater than KB, the result will always be positive.

In this scenario, two things could happen:

$\Delta AIC > 0$. If the value ΔAIC is positive, the improvement obtained in the fit of the curve measured in terms of differences in squares will not justify the increase in parameters.

$\Delta AIC < 0$. If the value ΔAIC is negative, the increase in the number of parameters will produce a better fit than forecasted, measured in terms of differences in squares.

Hence, if ΔAIC is negative, the model with more parameters is the one we will consider most valid in terms of fit.

	PCs (U.S.)	CD players (U.S.)	Internet hosts	Refrigerators (ES)
ΔAIC	-83,162	-0,877	-5,051	-101,112

Table 8. ΔAIC results. Proposed model compared to Bass model

As shown in Table 8, the ΔAIC values are negative in all cases, indicating that the proposed model provides the best fit.

The probability that a model will be correct can be expressed by the following indicator:

$$ER \text{ (Evidence ratio)} = 1/\exp(-0.5 \cdot \Delta AIC). \quad (22)$$

This is interpreted as follows: if it takes a value such as 10, for instance, the model with the lower AIC will be approximately 10 times more likely. The following table describes the Evidence Ratio values for the four products considered:

	PCs (U.S.)	CD players (U.S.)	Internet hosts	Refrigerators (ES)
Evidence Ratio	1.14366E+18	1.55E+00	1.25E+01	9.04107E+21

Table 9. Evidence Ratio of the proposed model compared to the Bass model

From the values in the table above, we can infer that the capacity of fit of the proposed model is significantly superior to the Bass model.

Graphical analysis

For a graphical view of the dual market, we have used the data obtained by solving the equation of Bass (14 and 16) with the coefficients from Table 4 for

each of the two markets. The results confirm the existence of a dual market in the four time series' on the adoption of new products (computers, CD players, Internet hosts and refrigerators).

The graphs of the non-cumulative annual growth of the four product families were also obtained for the Bass model and the proposed model, and were compared to the original data.

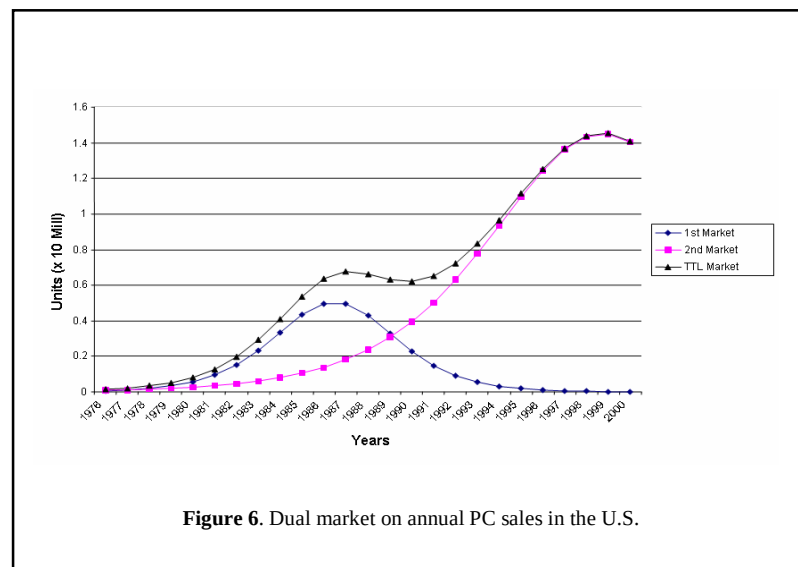
The purpose of this analysis is to graphically display and compare the fit of the data obtained with the proposed model to that of the classical Bass model. In all cases, the results show a superior fit, especially in areas of regression in the sales curve.

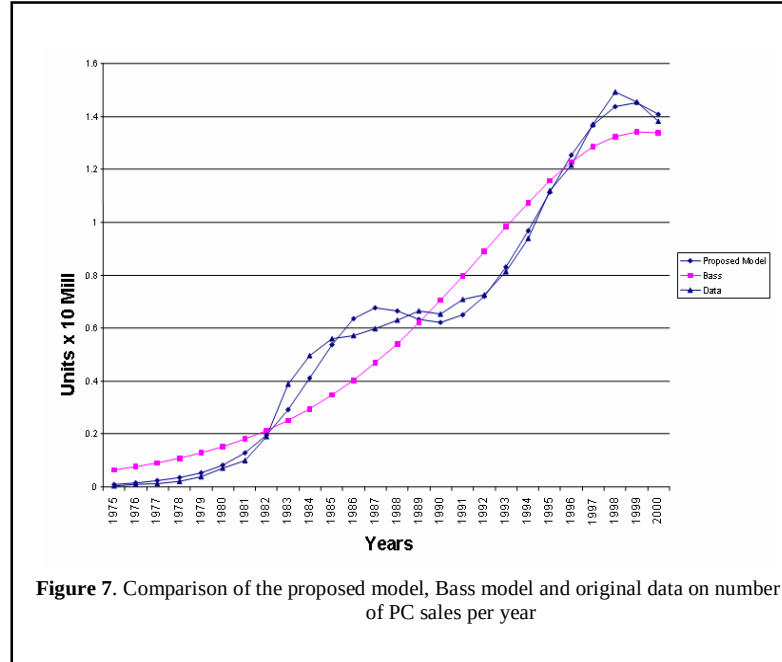
Case 1. PCs

The data reveal the existence of a dual market in which the sales peak for the first market occurs in the 11th year following introduction of the goods (Figure 6).

The data refer to all models of personal computers since the introduction of the Apple II in 1977. This was later joined by models from Commodore, Texas Instruments, Sinclair, Atari, Apple Macintosh and others. The peak in the first stage of the market also coincides with the launch of IBM PCs in 1981 and, later, of the IBP XT in 1983, the AT in 1984 and the 286 in 1986.

A graphical comparison of the sales data of the two models with the original data shows that the proposed model has greater forecasting ability than the Bass model (Figure 7).





Case 2. Number of Internet hosts

As Tehan (2002) notes, it is extremely difficult to estimate the number of users of the Internet, so its growth is usually estimated with another set of indirect indicators. One indirect indicator used to study the penetration and diffusion process of the Internet is the number of available hosts. López, Orero and Arroyo (2007) provide a development in the number of hosts from the birth of the Internet to 2006, which we will use for our analysis. For practical reasons, in this paper we have omitted data before 1991 because there are no time values with the same frequency and because the end result would not affect the purpose of our study since the area of regression occurs in 2000-2003 (Figure 9).

Figure 8 reveals the existence of dual markets in the number of Internet hosts and 2003 can be regarded as the point of intersection between the two markets. Although the difference between the two models is not significant (R^2 of 99.63% with the Bass model versus 99.82%), in the cumulative data on adoption, the graphical analysis shows that the model provides a better fit, especially in the areas of regression.

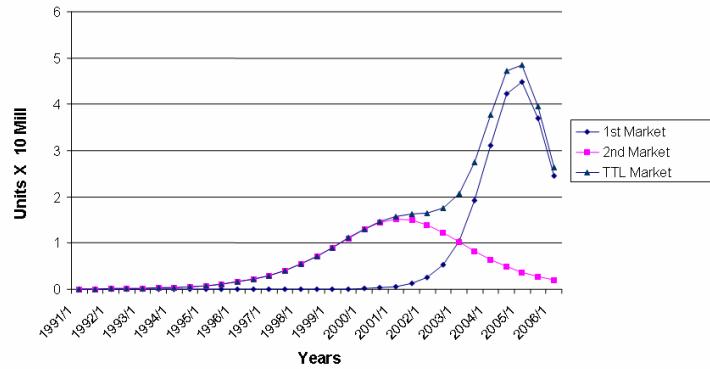


Figure 8. Dual markets of Internet hosts

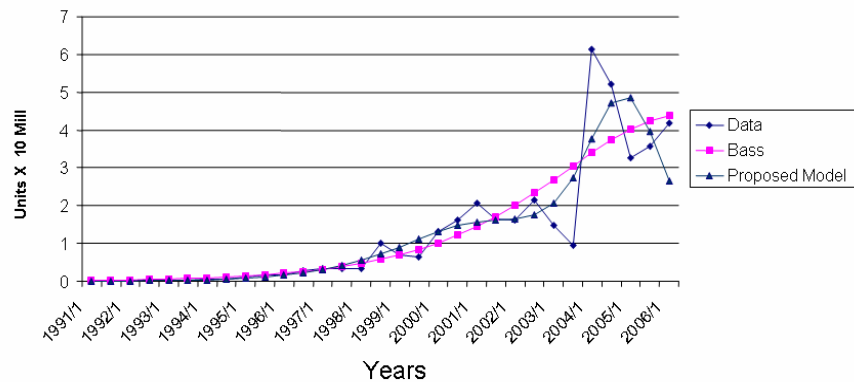


Figure 9. Comparison of the proposed model, Bass model and original data on number of Internet hosts per year

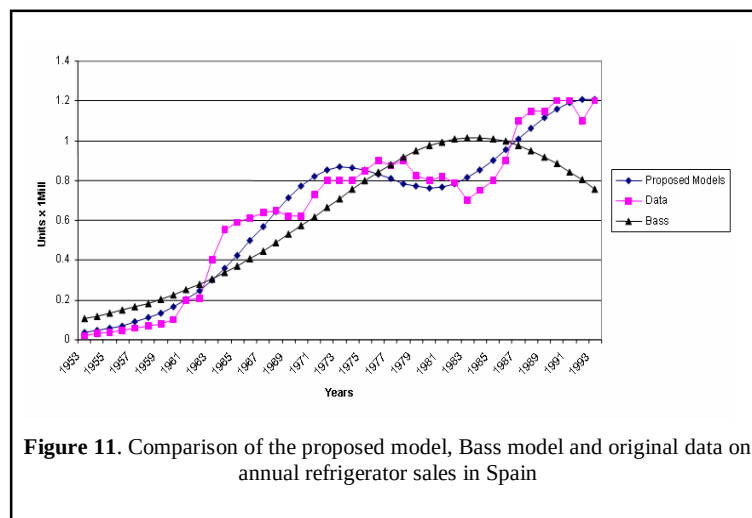
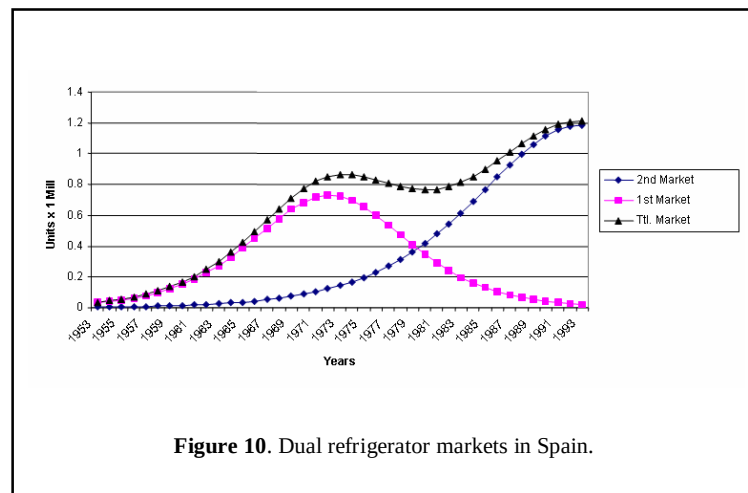
Case 3. Refrigerators in Spain

In the case of refrigerators, the first peak in sales is observed 20 years after their introduction. Of all the cases studied, this is the only pre-digital-generation product whose time to reach the first peak is exceptionally higher than that described by Golder and Tellis (2004), where the average was approximately 13 years.

In this case, we can distinguish between the three periods that shape the "saddle" effect. The first, up to 1973, is the initial growth period. This is followed by a

second period in which sales remain steady or decline over a period of more than 10 years before finally entering the third period, again of growth. These three stages are easily observed in the proposed model, which offers a better fit than the Bass model with the original data, especially in the area of regression (Fig. 11).

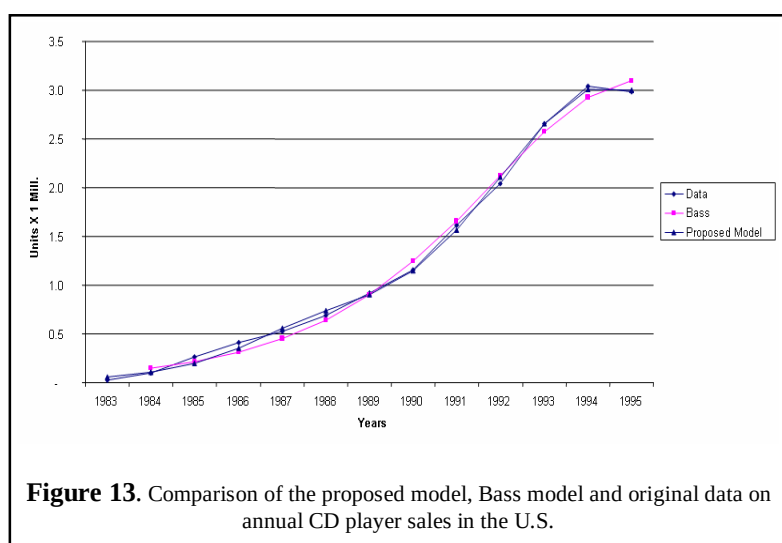
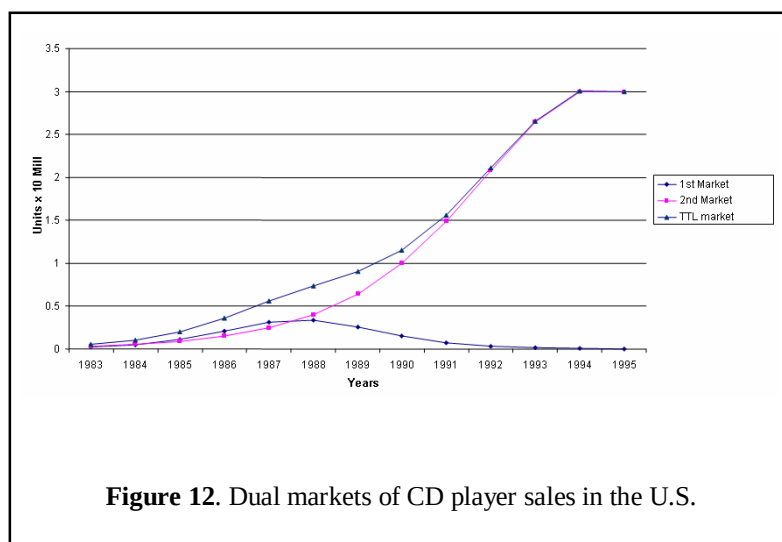
The graphical analysis and its results show that the proposed model offers a better approximation for analysing the diffusion of products in the presence of regressions in the adoption curve.



Case 4. CD players in the U.S.

CD players reach their first peak four years after their introduction, indicating exceptionally high growth. Annual sales of CD players have an almost imperceptible saddle effect. This case is particularly useful for analysing the behaviour of the model when no regression is observed in the sales curve.

As shown in Figure 12, the second market quickly reaches and overtakes the first, indicating a much faster diffusion process than the other technologies we have seen. Moreover, the differences between the Bass model and the proposed model are minimal and in both cases the fit is over 99.9%. This explains the minimal differences between the original data, the Bass model and the proposed model seen in the graph in Figure 13.



Conclusions

In this paper, we have proposed a diffusion model based on the classical Bass model (1969), which empirically explains the existence of a dual market in the adoption of new products. The mathematical model developed incorporates the proposal of two separate and unrelated markets in technology adoption previously pointed out conceptually by Moore (1991).

To validate the model, we used data from four types of product from different countries and technologies. The results show that its ability to predict the growth of these technologies is significantly superior to the classical Bass model and evidence was provided regarding the existence of the two markets in the four examples used.

Academically, this is a relevant result because the proposed model significantly improves the results obtained with the Bass model and helps bring a new perspective to diffusion studies by suggesting a revision based on the existence of two markets in adoption cycles.

It remains to be seen whether the model could be used to predict future sales and whether it could be extended with marketing variables. Its comparison to other diffusion models based on dual markets is also pending.

Limitations

In our proposal, we have tried to keep to the minimum level of complexity allowed by the model while also obtaining interesting and significant results. This produces certain limitations in its applicability. For example, the model is only valid for newly purchased products and does not include repeat product purchases. It is therefore limited to describing the adoption of new products or technologies where there are no subsequent purchases or where these occur in the very long term. As a result, there will be a single purchase per adopter in the time period defined by the model.

In addition, the model does not include interactions between different products or product categories that affect the diffusion process. It does not contain explanatory variables that could account for the influence of other, similar products on penetration rates. It also assumes a fixed and finite market size that can be known or estimated. The influence of exogenous factors, such as economic, social or technological variables, or endogenous factors, such as different generations or product enhancements, advertising or changes in distribution channels do not affect market size in the proposed model.

Lastly, the impact of marketing variables on the diffusion process are implicitly described through the coefficients of internal and external influence (p , q) and no specific effects of price, distribution or advertising are proposed in the diffusion model.

Despite the limitations of the proposed model, assumed for the sake of simplicity and interpretability, we believe that the empirical results of its validation and its comparison to alternative models show that it is a relevant contribution to the literature on innovation diffusion.

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